

## 5 Information transition

We have shown that an impurity moving in a 1D medium can exhibit a phase transition in operator Markovianity by studying the OTOC dynamics of the system with the impurity reset operation shown in Fig. 4. Operator dynamics has been linked to the spread of information [1, 14], and we will now explore how the phase transition in operator non-Markovianity manifests in the dynamics of quantum information. Specifically we study the coherent information transmitted through two complementary channels. These channels are those highlighted in the right-hand panel of Figure 8 as Yellow  $\rightarrow$  Blue for the echo protocol and Yellow  $\rightarrow$  Green for the Teleportation protocol. The results in this section were numerically simulated using the `QuantumClifford.jl` software package [90].

### 5.1 Echo protocol

In the thought experiment, Alice prepares the impurity qubit,  $I$ , in a particular state,  $\rho_I$ , and then allows it to move through the chaotic medium,  $M$ , evolving under the circuit described in Section 2. At an intermediate time she discards her qubit and replaces it with a maximally mixed qubit,  $R$ . At the end of the experiment, a second party, Bob, obtains access to both the discarded qubit,  $D$ , and the medium  $M$ . For clarity, we emphasise that Bob cannot access the impurity and has no knowledge of the input states.

We now show that the transition in operator backflow can also be observed in the information Bob has about the state,  $\rho_I$ , that Alice prepared in the impurity qubit at the beginning of the experiment. We quantify this information using the coherent information [91]. The coherent information is most easily calculated by introducing an ancilla qubit,  $A$ , that is maximally entangled with the initial state of Alice’s qubit (the impurity). Conceptually, this ancilla acts a memory of the initial state of the impurity, and allows us to track how much of that information remains on the impurity as it interacts with the medium. Note that this protocol is complementary to the one in which Alice prepares a specific state  $\rho_I$  on the impurity. The coherent information accessible by Bob is then given [91] as

$$I(A \rightarrow \text{Bob}) = S(\rho_{\text{Bob}}) - S(\rho_{\text{Bob} \cup A}), \quad (10)$$

where  $S(\rho_C) = -\text{Tr}[\rho_C \log_2 \rho_C]$  is the von-Neumann entanglement entropy of the reduced density matrix of subspace  $C$ . The subspace labelled “Bob” contains the qubits that Bob can access, namely the qubit discarded by Alice and the final state of the medium:  $\text{Bob} \equiv D \cup M$ . For a circuit which allows Bob to recover Alice’s state  $\rho_I$  perfectly, his qubits will end up maximally entangled with the ancilla  $A$  such that  $I(A \rightarrow \text{Bob}) = 1$ . While if Bob ends up with no information about Alice’s state, his qubits and the ancilla will decouple, and  $I(A \rightarrow \text{Bob}) \leq 0$ .

During the first stage of the protocol, the initial state is entirely encoded in the impurity and the medium subspaces. Immediately after this impurity is discarded, prior to the occurrence of any swaps, this is still true and Bob has perfect access to Alice’s initial qubit such that  $I(A \rightarrow \text{Bob}) = 1$ . As long as this remains true, Bob will always have complete information on the initial state. This is certainly true if the dynamics are Markovian. Here there is no operator back flow, the information remains in the medium and again Bob has perfect access to the information in Alice’s initial qubit. Once the operator dynamics become non-Markovian, however, some information flows back into Alice’s impurity and Bob loses the ability to decode the initial state of Alice’s qubits. This is shown in the left-hand panel of Figure 8 where we plot  $\overline{I(A \rightarrow \text{Bob})}$  as a function of  $v_d$ . There we see a transition occurring at the same critical velocity as the backflow transition observed by the OTOC. We again find the diffusive scaling exponents as shown in Figure 9, with the